Welcome to Your
Advanced Diploma in Generative AI

Build Smarter with LLMs, Prompting, Fine-tuning and RAG

COURSE WELCOME E-KIT

This e-kit gives you everything you need before your first class: course roadmap, learning outcomes, preparation checklist, project expectations, attendance tips, and support details.

Mode	Classroom + Online support available
Best For	Working Professionals
Outcome	Build confidence in LLM fundamentals, prompt engineering, fine-tuning, and RAG-based application development.

A Warm Welcome

Dear Learner,

Welcome to Expert Computer Academy. You have taken a smart step toward building practical, future-ready Generative AI skills. This course is designed to help you understand how modern AI systems work and how to apply them in real-world workflows, products, and business use cases.

By the end of this course, you should be able to understand LLM foundations, write better prompts, explore model customization, and build retrieval-augmented applications with confidence.

What You Will Gain

- Confidence in Generative AI concepts and LLM fundamentals
- Practical prompt engineering skills for quality outputs
- Exposure to fine-tuning workflows and model adaptation
- Ability to build RAG-based applications using real content
- A stronger professional profile for AI, product, operations, and automation-focused roles

> **Generative AI is not just a trend — it is a practical skill that helps professionals automate work, build smarter solutions, and stay ahead in a fast-changing digital world.**

Course Curriculum Roadmap

Module	Topic & Description
Module 1	**Foundations of LLMs** Tokens, parameters, context windows, embeddings, model behavior, use cases, limitations, and safe AI usage basics.
Module 2	**Prompt Engineering** Zero-shot and few-shot prompting, instruction design, role prompting, structured outputs, prompt templates, and prompt evaluation.
Module 3	**Fine-tuning** Fine-tuning concepts, dataset preparation, instruction tuning basics, parameter-efficient methods, and model improvement workflows.
Module 4	**RAG Apps** Embeddings, retrieval pipelines, vector stores, context injection, document Q&A; workflows, and basic RAG application building.

Practical Focus: Every topic should be practiced using hands-on exercises, simple datasets, real business use cases, and mini projects so learners can apply the concepts immediately.

Segment: Working Professionals

Learning Outcomes

1. Understand LLM Concepts Learn the core ideas behind how large language models generate and process information.	**2. Write Better Prompts** Use prompt structures that improve clarity, relevance, and response quality.
3. Customize AI Workflows Explore how fine-tuning and structured prompting adapt models for specific use cases.	**4. Build RAG Applications** Connect models with your own documents and data for grounded answers.
5. Work with AI Tools Understand common tools used in GenAI experimentation, testing, and deployment.	**6. Prepare for Projects & Interviews** Discuss GenAI concepts, use cases, and implementation ideas with confidence.

Mini Portfolio Projects You Can Build

- AI Content Assistant
- Document Q&A; Bot
- FAQ / Knowledge Base Assistant
- Prompt Library for Business Tasks
- Simple RAG-based Chatbot

Before Your First Class

Use this checklist to come prepared and get maximum value from every session.

- Install Python 3.11+ and a code editor such as VS Code or Jupyter Notebook.
- Create accounts for OpenAI, Hugging Face, or other tools suggested by the instructor.
- Install Git and basic required Python packages if instructed.
- Create a folder named Generative AI Practice to save notes, prompts, and mini projects.
- Keep a notebook or digital notes app for concepts, prompts, and implementation steps.
- Read a basic introduction to terms like LLM, prompt, token, embedding, and RAG before class.

Recommended Learning Discipline

Focus Area	Recommended Practice
Daily practice	30 minutes after every class
Weekly revision	Review prompts, concepts, and examples once a week
Project habit	Save every class task as a mini portfolio sample
Doubt clearing	Ask doubts before moving to the next module

Class Guidelines & Support

During Class	After Class
• Join on time and keep your notebook or coding setup ready • Follow the instructor step by step before experimenting further • Do not skip hands-on practice activities • Ask questions whenever a concept or output is unclear	• Complete the given practice activity or mini exercise • Save your work with proper file names and notes • Revise the concept behind the output, not just the result • Share doubts with screenshots, prompts, or code references

Support Details

Phone	7282983335 | 9128812325
Address	Kumar Tower, 2nd Floor, Boring Rd Crossing, Patna-800001, Bihar
Online Course	Online courses also available
Academy	Expert Computer Academy - 38 Years of Excellence

Starter Generative AI Cheat Sheet

Keep this page handy during the first few sessions. The instructor will explain these terms with practice examples.

Term	Meaning	Example / Use
LLM	Large language model used to understand and generate text	Example: GPT-style assistants
Prompt	Instruction or input given to the model	Example: “Summarize this email professionally”
Temperature	Controls how creative or predictable the response is	Lower for factual tasks, higher for creative tasks
Token	Small unit of text processed by the model	Affects cost and context length
Embedding	Numerical representation of text for semantic search	Used in retrieval systems
Fine-tuning	Adapting a model with task-specific examples	Useful for specialized behavior
RAG	Retrieval-Augmented Generation using external knowledge	Used for document Q&A;
Vector DB	Database that stores embeddings for retrieval	Used with RAG pipelines

Tip: Do not memorize terms blindly. Understand what problem each concept solves, then apply it through small hands-on experiments.

Your 30-Day Generative AI Growth Plan

Timeline	Focus Area
Week 1	LLM basics + core concepts + prompt fundamentals
Week 2	Advanced prompting + structured outputs + evaluation practice
Week 3	Fine-tuning concepts + dataset thinking + model customization basics
Week 4	RAG pipeline + mini project + interview revision

Final Message

Generative AI becomes powerful when you practice consistently. Attend each session, complete your exercises, and build your own mini project. Your goal is not only to finish the course — your goal is to become confident enough to apply AI in real work.

Ready to begin? Save this e-kit, attend your first class, and start building your Generative AI confidence from Day 1.